

# DINESH KUMAR M

Software Engineer | Full Stack Engineer (Backend Focused)

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## PROFESSIONAL SUMMARY

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Backend-focused Full-Stack Software Engineer with **2+ years of professional experience** building and supporting production applications serving **20K+ registered users**. Experienced in **Java, Rust, Spring Boot, PostgreSQL, DynamoDB, Redis, REST, GraphQL, gRPC, AWS, microservices, and distributed systems**, with production frontend experience in **Next.js and Angular**. Contributed to feature-level architecture, API and database design, asynchronous workflows, deployment validation, and production troubleshooting across 25+ releases.

## TECHNICAL SKILLS

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**Languages:** Java, Rust, TypeScript, JavaScript, SQL

**Backend:** Spring Framework, Spring Boot, Spring MVC, Spring Data JPA, Axum, Tokio, Tonic, NestJS, Node.js

**Frontend:** Next.js, Angular, React

**APIs & Integration:** REST APIs, GraphQL, Apollo GraphQL, gRPC, Protocol Buffers, Third-Party API Integration

**Databases & Caching:** PostgreSQL, MySQL, DynamoDB, Redis, ElastiCache/Valkey

**Architecture & Backend Engineering:** Microservices, Distributed Systems, Monolithic Architecture, API Gateway, API Design, Data Modeling, Asynchronous Workflows, Authentication & Session Workflows, RBAC, Transactional Workflows

**Cloud, Messaging & DevOps:** AWS, AWS SQS, Docker, ECR, EKS Deployment, CloudWatch, Jenkins, ArgoCD, Grafana

**Developer Tools:** Git, Maven, Cargo, Postman, Altair GraphQL, RedisInsight, pgAdmin, NoSQL Workbench

## PROFESSIONAL EXPERIENCE

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**Aretedge Innovations Pvt. Ltd.**

Associate Software Developer

Hyderabad, Telangana, India (On-site)

Aug 2024 – May 2026

- Owned **90% of the Challenge backend** and **90% of Casual Games**, covering API/database design, business logic, integrations, deployment validation, and production support.
- Developed production backend functionality across **Java/Spring Boot and Rust-based microservices**, implementing APIs, service-layer business logic, asynchronous workflows, database interactions, and service integrations.
- Developed and modified **20–30 production APIs** using **REST, GraphQL, and gRPC** across Rust, Java, and NestJS backend services.
- Created and modified **Protocol Buffer contracts**, including service definitions, request/response messages, and enums, and implemented corresponding backend integrations using gRPC.
- Contributed approximately **50% to the internal GraphQL API Gateway**, implementing schemas, models, resolvers/handlers, and service integrations connecting client-facing GraphQL operations with internal gRPC services.
- Worked within an **Axum, Tokio, and async-graphql Gateway architecture** supporting asynchronous HTTP/gRPC workflows and a common API boundary across distributed backend services.
- Enhanced existing authentication workflows involving **OTP, JWT access tokens, refresh-token handling/rotation**, and Redis-backed temporary session state in distributed backend services.
- Created or significantly modified **6+ production PostgreSQL tables and 12–15 functions/stored procedures**, using JSONB and materialized views for transactional and leaderboard/ELO workloads.
- Supported migration of approximately **20K user/login records from DynamoDB to PostgreSQL** using Node.js when batch/filter-heavy access patterns became inefficient for the workload.
- Used **Redis/ElastiCache** for authentication/session state and leaderboard workloads spanning daily, weekly, monthly, and country-based rankings across approximately **10–20 country variations**.
- Implemented end-to-end **Double Elimination tournament logic**, including winners/losers brackets, match progression, participant advancement, finals, and winner determination.
- Worked with **AWS SQS-based asynchronous application workflows** and AWS-hosted services including Aurora PostgreSQL, DynamoDB, ElastiCache/Valkey, ECR, EKS-hosted applications, and CloudWatch.

- Independently handled approximately **4 Next.js applications** in a production white-labeled monorepo and developed Angular pages/components integrated with Java/Spring Boot APIs.
- Supported **25+ production releases, 10+ incidents, and 8+ hotfixes** through log analysis, root-cause investigation, API validation, Jenkins deployments, ArgoCD/Grafana monitoring, and production verification.

### Aspire Systems

Software Engineering Intern

Chennai, Tamil Nadu, India (Remote)

Jan 2023 – Mar 2023

- Completed project-based software engineering training using **Java, MySQL, and Angular**, applying object-oriented programming, relational database operations, frontend development, debugging, Git, and application testing.

## PROJECTS

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**Personal Finance Tracker** | *Next.js, TypeScript, PostgreSQL, NeonDB, Zustand*

**Links:** [Live Demo](#) | [GitHub](#)

- Built and deployed a full-stack personal finance application with authentication, transaction tracking, analytics, monthly/yearly reporting, CSV import, and PostgreSQL-backed data workflows.
- Reworked the data layer from **Supabase + Drizzle ORM to Neon PostgreSQL**, simplifying the application's database setup while retaining Row-Level Security and core financial-data workflows.

**Student Management System** | *Java, Angular 15, MySQL, JSON Server*

**Links:** [Live Demo](#) | [GitHub \(Angular\)](#) | [GitHub \(Java\)](#)

- Built a **Java CLI-based Student Management System with MySQL**, implementing student-management workflows using object-oriented programming and relational database operations.
- Developed a separate **Angular 15 web application** for academic workflows using components, services, HttpClient, forms, routing, and a JSON Server mock REST backend.

## EDUCATION

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**K.S.R. College of Engineering**

*Bachelor of Engineering (B.E.) in Computer Science and Engineering* | **CGPA: 8.56/10**

Namakkal, Tamil Nadu, India

Aug 2019 – May 2023

## CERTIFICATIONS

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**Rust Programming Essentials** – Edureka

**AWS DynamoDB Fundamentals** – Meta

**Advanced React** – Whizlabs

**Google Prompting Essentials** – Google